Hydrogen gas, 2 gm. litre;
Oxygen gas, 32 gm. litre;
Water vapour, 18 gm. litre.

Accordingly, the relative masses of the various molecules in question are:

Hydrogen gas: oxygen gas: water vapour = 2: 32: 18

The hypothesis which expresses these facts most simply, and which has been confirmed by other experiments, is that a hydrogen molecule consists of two hydrogen atoms of atomic weight 1, and consequently, has the molecular weight 2. Similarly, an oxygen molecule consists of two oxygen atoms, of atomic weight 16, and consequently, has the molecular weight 32. And, finally, a water molecule consists of two hydrogen atoms and one oxygen atom, and consequently, has the molecular weight $2 \times 1 + 16 = 18$. This compound can be described by the following formula:

$$1\ H_2 + \tfrac{1}{2}O_2 = 1\ H_2O$$

The atomic weights of all elements, as well as the molecular weights of all chemical compounds, can be determined in analogous manner. The unit of atomic weight is chosen so as to be equal exactly to $\frac{1}{16}$ of the chemical atomic weight of oxygen, so that the atomic weight of oxygen is exactly 16.00000. Similarly, the molecular weight is the standard measure of the mass of a molecule measured in terms of the same unit.

We have already mentioned the unit called *mol* (or *mole*). One mol represents that quantity of a substance which weighs as many grammes as equal numerically the molecular weight of that substance. Thus, one mol of hydrogen gas (H_2) is 2 grammes, one mol of oxygen gas (O_2) is 32 grammes, and 1 mol of water (H_2O) is 18 grammes. Thus the masses of 1 mol of different substances are to each other exactly as their respective molecular weights, and consequently, as the masses of their individual molecules. It follows, therefore, that 1 mol of any substance always contains exactly the same number of molecules. Quite analogously, a gramme-atom is defined as that quantity of an element which weighs as many grammes as equal